TCS349: Responsible and Explainable AI

Semester-Wise Lecture Plan

Programme: B.Tech. Computer Science and Engineering
Semester: III
Credits: 3
Total Teaching Hours: 42
Teaching Load: Approximately 3 lectures per week

The plan follows the five-unit syllabus and prescribed teaching hours: Unit I-9 hours, Unit II-9 hours, Unit III-7 hours, Unit IV-9 hours, and Unit V-8 hours.

Unit I: Introduction to AI Ethics, Bias and Fairness

Teaching Hours: 9
Course Outcome: CO1

Lecture	Topic to be Covered	Teaching-Learning Method
1	Introduction to AI ethics, Responsible AI and the importance of ethical AI development	Lecture and real-world examples
2	Ethics, morality and law; fundamental ethical principles for AI systems	Lecture and classroom discussion
3	Common ethical dilemmas in AI and stakeholder-oriented ethical analysis	Case-based discussion
4	Bias in AI systems: historical, data, sampling, measurement and label bias	Lecture and illustrative examples
5	Algorithmic bias, interactional bias, feedback loops and deployment bias	Case analysis and discussion
6	Detecting and measuring bias using subgroup analysis and confusion-matrix-based measures	Numerical examples and demonstration
7	Fairness in AI: group fairness, individual fairness and demographic parity	Conceptual explanation and examples
8	Fairness metrics and trade-offs: equal opportunity, equalized odds and predictive parity	Problem-solving and comparison
9	Bias mitigation strategies; imminent risks and long-term risks from AI models	Framework discussion and case study

Self-Learning Activity: Review a real AI failure and identify the ethical issue, source of bias, affected stakeholders and relevant fairness concerns.
Team Activity: Analyse an AI application used in daily life and propose measures to make it fairer and safer.

Unit II: Transparency, Accountability, Surveillance and Privacy

Teaching Hours: 9
Course Outcome: CO2

Lecture	Topic to be Covered	Teaching-Learning Method
10	Principles of Responsible AI and the trustworthy AI lifecycle	Lecture and framework discussion
11	Transparency in AI: meaning, importance, traceability and communication of limitations	Lecture and examples
12	Transparency tools: model cards, datasheets, system cards, documentation and audit logs	Template-based demonstration
13	Accountability in AI: roles, responsibility, human oversight and escalation mechanisms	Case-based discussion
14	Accountability frameworks, organisational governance and regulatory compliance	Framework comparison
15	AI applications in surveillance: facial recognition, video analytics and behavioural monitoring	Lecture and case examples
16	Privacy concerns: data collection, consent, profiling, purpose limitation and data misuse	Classroom discussion
17	Data-protection principles and privacy-aware techniques: minimisation, anonymisation and access control	Demonstration and examples
18	Balancing security and privacy in AI implementations; unit case study and review	Debate and problem-solving

Self-Learning Activity: Prepare a transparency and privacy checklist for a selected AI system using model-card and data-documentation principles.
Team Activity: Debate the responsible use of facial recognition or AI-enabled surveillance in public spaces.

Unit III: Societal Impacts of AI

Teaching Hours: 7
Course Outcome: CO3

Lecture	Topic to be Covered	Teaching-Learning Method
19	AI and employment: task automation, job displacement and changing occupational structures	Lecture and labour-market examples
20	Job augmentation, human-AI collaboration and redesign of work	Scenario-based discussion
21	AI-driven productivity, economic growth and transformation of industries	Lecture and sectoral case studies
22	AI, inequality, concentration of power and the digital divide	Critical discussion
23	Emerging skill requirements, reskilling, upskilling and workforce transition	Activity and career mapping
24	AI in governance and public policy: public services, decision support and algorithmic administration	Policy case analysis
25	Long-term societal implications of AI: democracy, social trust, autonomy and human values	Seminar and reflective discussion

Self-Learning Activity: Write a short policy brief on how AI may transform one profession, industry or public service.
Team Activity: Map the positive and negative societal impacts of AI in education, healthcare, finance or governance.

Unit IV: Explainable Artificial Intelligence Fundamentals

Teaching Hours: 9
Course Outcome: CO4

Lecture	Topic to be Covered	Teaching-Learning Method
26	Introduction to Explainable AI and the need for explainability in AI systems	Lecture and motivating examples
27	Interpretability versus explainability; explanation stakeholders, goals and requirements	Comparative explanation
28	Taxonomy of XAI methods: intrinsic and post-hoc, local and global, model-specific and model-agnostic	Diagram-based explanation
29	Local explanations using LIME: concept, perturbation, local surrogate model and workflow	Algorithm demonstration
30	Implementing and interpreting LIME explanations using Python	Hands-on Python demonstration
31	SHAP fundamentals: Shapley values, feature attribution and explanation properties	Conceptual and numerical examples
32	Implementing and interpreting SHAP explanations using Python	Hands-on Python demonstration
33	Global explanations: feature importance, permutation importance and perturbation analysis	Demonstration and comparison
34	Decision trees as explanatory models, surrogate models and rule-based explanations	Rule extraction and comparative activity

Self-Learning Activity: Implement LIME, SHAP and feature-importance methods on a classification dataset and compare the explanations.
Team Activity: Critically compare local and global explanation techniques for the same prediction model.

Unit V: Visualization and Applications of Explainable AI

Teaching Hours: 8
Course Outcome: CO5

Lecture	Topic to be Covered	Teaching-Learning Method
35	Visualisation of feature effects and model decisions using heatmaps, PDP and ICE plots	Lecture and visual examples
36	Implementing Partial Dependence Plots and Individual Conditional Expectation plots in Python	Hands-on Python demonstration
37	Visualising local and global model decisions using SHAP summary, dependence and force-style plots	Demonstration and interpretation
38	Neural-network and layer-activation visualisation: saliency maps, activation maps and Grad-CAM concepts	Visual demonstration
39	XAI in finance: credit scoring, loan approval and fraud detection	Domain case study
40	XAI in healthcare: diagnostic systems and treatment recommendations	Domain case study
41	XAI in autonomous systems: vehicle-control systems, robotics and safety-critical decisions	Scenario analysis
42	Comparative XAI practical, team presentations, interpretation guidelines and course wrap-up	Team activity and presentation

Self-Learning Activity: Create and interpret PDP, ICE, SHAP or activation-based visualisations for a trained model using Python.
Team Activity: Formal two-hour group activity to propose a responsible and explainable solution for fraud detection, disease diagnosis or autonomous decision-making.

Semester-Wise Weekly Distribution

Semester Week	Lectures	Unit and Coverage
Week 1	1-3	Unit I: Responsible AI introduction, ethics, morality, law and ethical dilemmas
Week 2	4-6	Unit I: Types of bias, feedback loops, bias detection and measurement
Week 3	7-9	Unit I: Fairness definitions, metrics, mitigation and AI risks
Week 4	10-12	Unit II: Responsible AI principles, transparency and documentation tools
Week 5	13-15	Unit II: Accountability, governance, compliance and surveillance applications
Week 6	16-18	Unit II: Privacy, data protection and security-privacy balance
Week 7	19-21	Unit III: Employment, augmentation, productivity and economic transformation
Week 8	22-24	Unit III: Inequality, emerging skills, governance and public policy
Week 9	25-27	Unit III completion and Unit IV introduction, interpretability and stakeholders
Week 10	28-30	Unit IV: XAI taxonomy and LIME concept and implementation
Week 11	31-33	Unit IV: SHAP concept and implementation; global explanation techniques
Week 12	34-36	Unit IV completion and Unit V feature-effect visualisation, PDP and ICE
Week 13	37-39	Unit V: Model-decision visualisation, neural activations and finance applications
Week 14	40-42	Unit V: Healthcare, autonomous systems, team activity and course wrap-up

Suggested Assessment Distribution

- After Unit I: Quiz and case analysis on ethics, bias, fairness and AI risks
- After Unit II: Assignment to audit the transparency, accountability and privacy of an AI system
- After Unit III: Group discussion or short policy brief on the societal and economic impacts of AI
- After Unit IV: Python assignment implementing and comparing LIME, SHAP and feature importance
- After Unit V: XAI visualisation task, team presentation and semester revision